# COVID-19 Mexico — Exploratory Data Analysis (EDA)

**Internship:** CodeAlpha | **Task:** Task 2
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## Meaningful Questions Before Analysis

1. What is the overall death rate of COVID-19 patients in Mexico?
2. Does age play a significant role in patient mortality?
3. Which comorbidities (Diabetes, Hypertension etc.) increase death risk?
4. Is there a difference in death rate between male and female patients?
5. What percentage of patients were hospitalized vs ambulatory?
6. Are there any outliers or anomalies in the age data?
7. Is there missing or inconsistent data that needs to be handled?
8. Does pneumonia presence increase the risk of death?

```
# 1. Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import warnings
warnings.filterwarnings('ignore')
sns.set_theme(style='whitegrid')
print('Libraries imported!')
```

```
Libraries imported!
```

```
# 2. Load Dataset
df = pd.read_csv("E:\\Ineternship\\Task 2\\Covid Data.csv",
low_memory=False)
print('Shape:', df.shape)
df.head()
```

```
Shape: (1048575, 21)
```

```
   USMER  MEDICAL_UNIT  SEX  PATIENT_TYPE   DATE_DIED  INTUBED
PNEUMONIA  \
0      2             1    1             1  03/05/2020       97
1
1      2             1    2             1  03/06/2020       97
1
2      2             1    2             2  09/06/2020        1
2
```

```
3      2             1    1             1  12/06/2020       97
2
4      2             1    2             1  21/06/2020       97
2

   AGE  PREGNANT  DIABETES  ...  ASTHMA  INMSUPR  HIPERTENSION
OTHER_DISEASE  \
0   65         2         2  ...       2        2             1
2
1   72        97         2  ...       2        2             1
2
2   55        97         1  ...       2        2             2
2
3   53         2         2  ...       2        2             2
2
4   68        97         1  ...       2        2             1
2

   CARDIOVASCULAR  OBESITY  RENAL_CHRONIC  TOBACCO
CLASIFFICATION_FINAL  ICU
0               2        2              2        2
3   97
1               2        1              1        2
5   97
2               2        2              2        2
3    2
3               2        2              2        2
7   97
4               2        2              2        2
3   97

[5 rows x 21 columns]
```

```
# 3. Data Structure – Variables & Data Types
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1048575 entries, 0 to 1048574
Data columns (total 21 columns):
 #   Column                Non-Null Count    Dtype
---  ------                --------------    -----
 0   USMER                 1048575 non-null  int64
 1   MEDICAL_UNIT          1048575 non-null  int64
 2   SEX                   1048575 non-null  int64
 3   PATIENT_TYPE          1048575 non-null  int64
 4   DATE_DIED             1048575 non-null  object
 5   INTUBED               1048575 non-null  int64
 6   PNEUMONIA             1048575 non-null  int64
 7   AGE                   1048575 non-null  int64
 8   PREGNANT              1048575 non-null  int64
```

```
 9   DIABETES              1048575 non-null  int64
 10  COPD                  1048575 non-null  int64
 11  ASTHMA                1048575 non-null  int64
 12  INMSUPR               1048575 non-null  int64
 13  HIPERTENSION          1048575 non-null  int64
 14  OTHER_DISEASE         1048575 non-null  int64
 15  CARDIOVASCULAR        1048575 non-null  int64
 16  OBESITY               1048575 non-null  int64
 17  RENAL_CHRONIC         1048575 non-null  int64
 18  TOBACCO               1048575 non-null  int64
 19  CLASIFFICATION_FINAL  1048575 non-null  int64
 20  ICU                   1048575 non-null  int64
dtypes: int64(20), object(1)
memory usage: 168.0+ MB
```

```
# 4. Statistical Summary
df.describe()
```

```
              USMER  MEDICAL_UNIT           SEX  PATIENT_TYPE
INTUBED  \
count  1.048575e+06  1.048575e+06  1.048575e+06  1.048575e+06
1.048575e+06
mean   1.632194e+00  8.980565e+00  1.499259e+00  1.190765e+00
7.952288e+01
std    4.822084e-01  3.723278e+00  4.999997e-01  3.929041e-01
3.686889e+01
min    1.000000e+00  1.000000e+00  1.000000e+00  1.000000e+00
1.000000e+00
25%    1.000000e+00  4.000000e+00  1.000000e+00  1.000000e+00
9.700000e+01
50%    2.000000e+00  1.200000e+01  1.000000e+00  1.000000e+00
9.700000e+01
75%    2.000000e+00  1.200000e+01  2.000000e+00  1.000000e+00
9.700000e+01
max    2.000000e+00  1.300000e+01  2.000000e+00  2.000000e+00
9.900000e+01
```

```
          PNEUMONIA           AGE      PREGNANT      DIABETES
COPD  \
count  1.048575e+06  1.048575e+06  1.048575e+06  1.048575e+06
1.048575e+06
mean   3.346831e+00  4.179410e+01  4.976558e+01  2.186404e+00
2.260569e+00
std    1.191288e+01  1.690739e+01  4.751073e+01  5.424242e+00
5.132258e+00
min    1.000000e+00  0.000000e+00  1.000000e+00  1.000000e+00
1.000000e+00
25%    2.000000e+00  3.000000e+01  2.000000e+00  2.000000e+00
2.000000e+00
50%    2.000000e+00  4.000000e+01  9.700000e+01  2.000000e+00
```

```
2.000000e+00
75%    2.000000e+00  5.300000e+01  9.700000e+01  2.000000e+00
2.000000e+00
max    9.900000e+01  1.210000e+02  9.800000e+01  9.800000e+01
9.800000e+01

             ASTHMA       INMSUPR  HIPERTENSION  OTHER_DISEASE  \
count  1.048575e+06  1.048575e+06  1.048575e+06   1.048575e+06
mean   2.242626e+00  2.298132e+00  2.128989e+00   2.435143e+00
std    5.114089e+00  5.462843e+00  5.236397e+00   6.646676e+00
min    1.000000e+00  1.000000e+00  1.000000e+00   1.000000e+00
25%    2.000000e+00  2.000000e+00  2.000000e+00   2.000000e+00
50%    2.000000e+00  2.000000e+00  2.000000e+00   2.000000e+00
75%    2.000000e+00  2.000000e+00  2.000000e+00   2.000000e+00
max    9.800000e+01  9.800000e+01  9.800000e+01   9.800000e+01

       CARDIOVASCULAR       OBESITY  RENAL_CHRONIC       TOBACCO  \
count    1.048575e+06  1.048575e+06   1.048575e+06  1.048575e+06
mean     2.261810e+00  2.125176e+00   2.257180e+00  2.214333e+00
std      5.194850e+00  5.175445e+00   5.135354e+00  5.323097e+00
min      1.000000e+00  1.000000e+00   1.000000e+00  1.000000e+00
25%      2.000000e+00  2.000000e+00   2.000000e+00  2.000000e+00
50%      2.000000e+00  2.000000e+00   2.000000e+00  2.000000e+00
75%      2.000000e+00  2.000000e+00   2.000000e+00  2.000000e+00
max      9.800000e+01  9.800000e+01   9.800000e+01  9.800000e+01

       CLASIFFICATION_FINAL           ICU
count          1.048575e+06  1.048575e+06
mean           5.305653e+00  7.955397e+01
std            1.881165e+00  3.682307e+01
min            1.000000e+00  1.000000e+00
25%            3.000000e+00  9.700000e+01
50%            6.000000e+00  9.700000e+01
75%            7.000000e+00  9.700000e+01
max            7.000000e+00  9.900000e+01
```

```python
# 5. Missing Values
missing = df.isnull().sum()
missing = missing[missing > 0]
print('Missing Values:\n', missing)
```

```
Missing Values:
 Series([], dtype: int64)
```

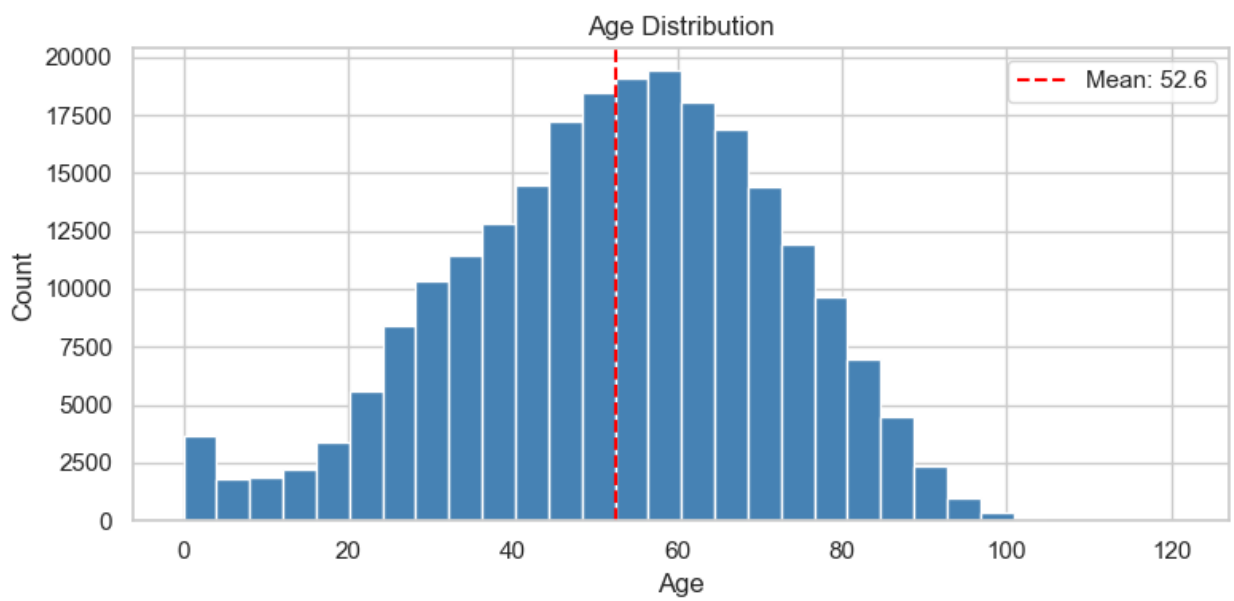
```python
# 6. Duplicate Rows
print('Duplicates:', df.duplicated().sum())
df = df.drop_duplicates()
print('After removing duplicates:', df.shape)
```

```
Duplicates: 812049
After removing duplicates: (236526, 21)
```

```python
# 7. Feature Engineering
df['DATE_DIED'] = pd.to_datetime(df['DATE_DIED'], errors='coerce')

df['DIED']      = df['DATE_DIED'].notna().astype(int)
df['GENDER']    = df['SEX'].map({1: 'Female', 2: 'Male'})
df['AGE_GROUP'] = pd.cut(df['AGE'], bins=[0,18,30,45,60,75,120],
                         labels=['0-18','19-30','31-45','46-60','61-
75','75+'],
                         right=False)

print('Death Rate:', round(df['DIED'].mean() * 100, 2), '%')
```

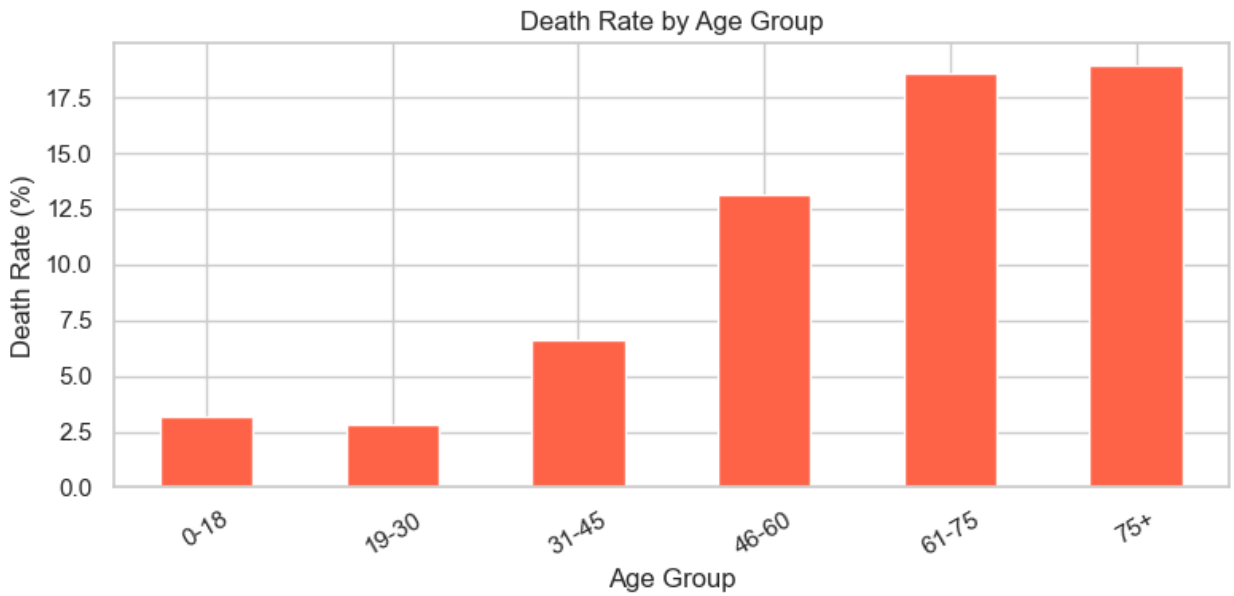
```
Death Rate: 12.73 %
```

```python
# 8. Age Distribution
plt.figure(figsize=(8, 4))
plt.hist(df['AGE'].dropna(), bins=30, color='steelblue',
edgecolor='white')
plt.axvline(df['AGE'].mean(), color='red', linestyle='--',
label=f'Mean: {df["AGE"].mean():.1f}')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')
plt.legend()
plt.tight_layout()
plt.show()
print('Most patients are between 30-60 years old.')
```

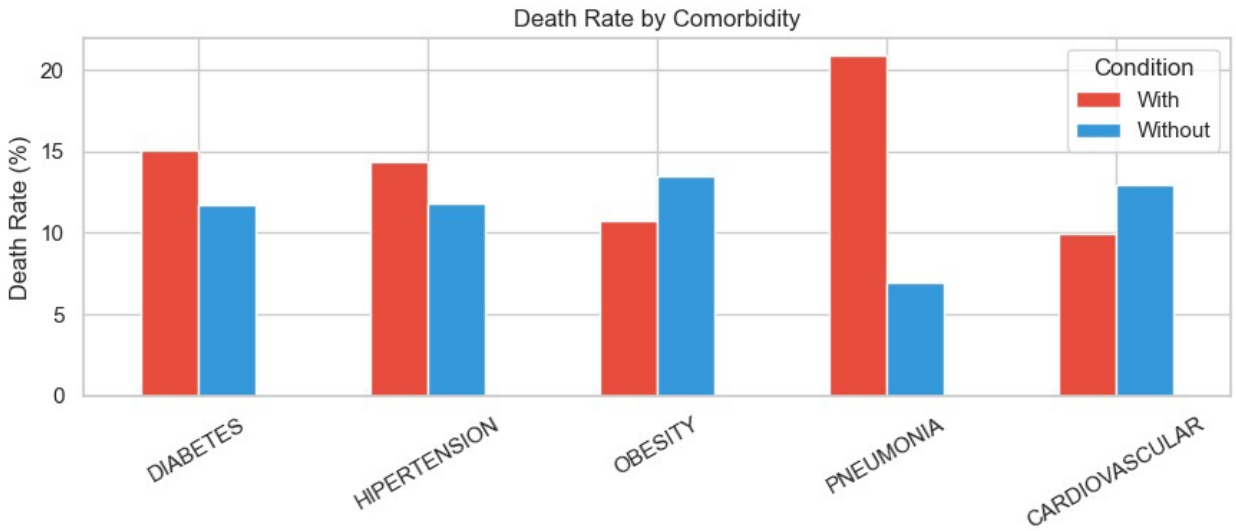


```
Most patients are between 30-60 years old.

# 9. Gender Distribution
gc = df['GENDER'].value_counts()
plt.figure(figsize=(6, 4))
plt.bar(gc.index, gc.values, color=['#FF9999','#66B2FF'],
edgecolor='white')
plt.title('Gender Distribution')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```

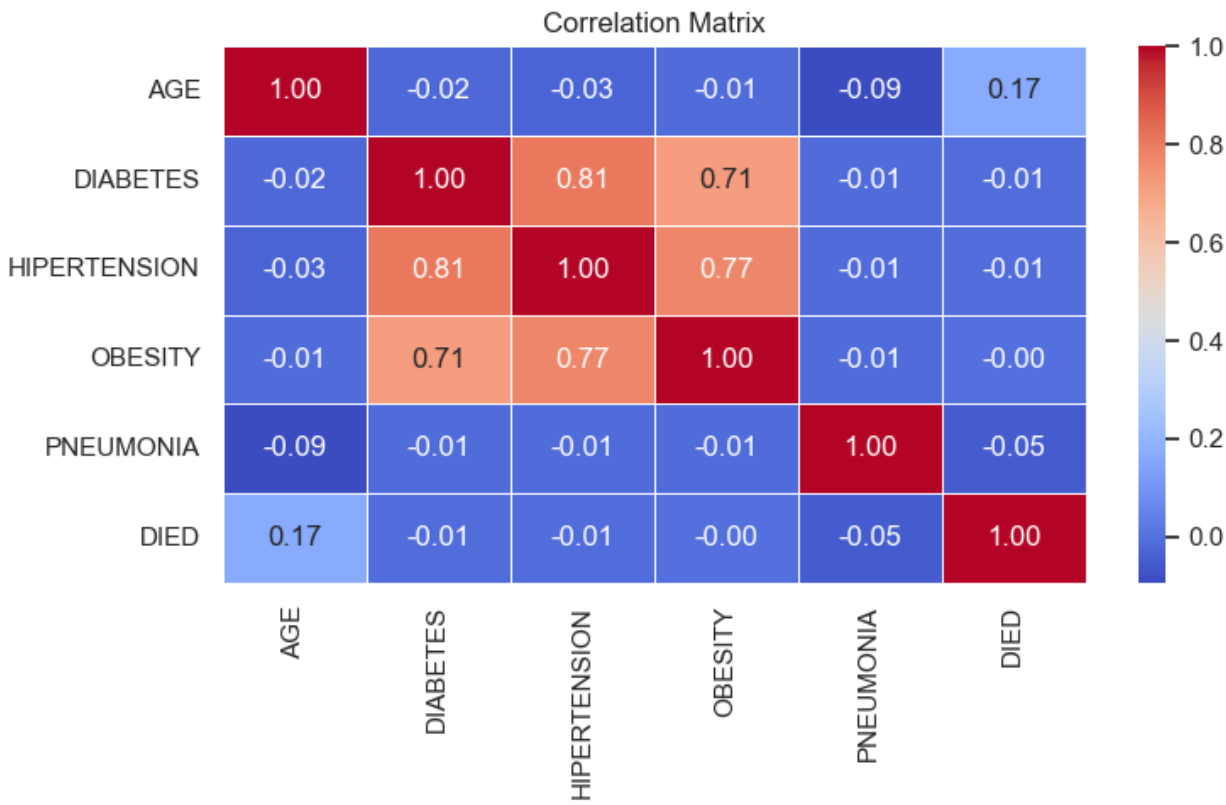
```
# 10. Death Rate by Age Group
death_by_age = df.groupby('AGE_GROUP', observed=True)['DIED'].mean() *
100
plt.figure(figsize=(8, 4))
death_by_age.plot(kind='bar', color='tomato', edgecolor='white')
plt.title('Death Rate by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Death Rate (%)')
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
print('Older patients (60+) have significantly higher death rate.')
```




```
Older patients (60+) have significantly higher death rate.

# 11. Comorbidities vs Death Rate
comorbidities =
['DIABETES','HIPERTENSION','OBESITY','PNEUMONIA','CARDIOVASCULAR']
available = [c for c in comorbidities if c in df.columns]

rates = {c: {'With'    : df[df[c]==1]['DIED'].mean()*100,
             'Without': df[df[c]==2]['DIED'].mean()*100} for c in
available}

pd.DataFrame(rates).T.plot(kind='bar', figsize=(9, 4),
                           color=['#E74C3C','#3498DB'],
edgecolor='white')
plt.title('Death Rate by Comorbidity')
plt.ylabel('Death Rate (%)')
plt.xticks(rotation=30)
plt.legend(title='Condition')
plt.tight_layout()
plt.show()
```

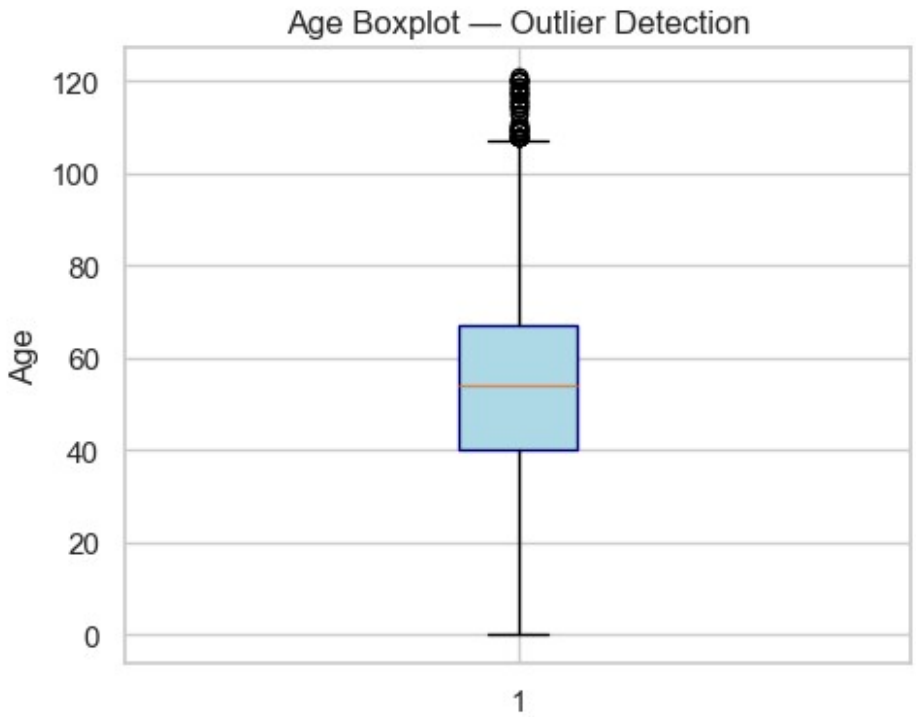



```
# 12. Correlation Heatmap
cols = [c for c in
['AGE','DIABETES','HIPERTENSION','OBESITY','PNEUMONIA','DIED'] if c in
df.columns]
plt.figure(figsize=(8, 5))
sns.heatmap(df[cols].corr(), annot=True, fmt='.2f', cmap='coolwarm',
linewidths=0.5)
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()
```




```python
# 13. Outlier Detection (IQR)
Q1, Q3 = df['AGE'].quantile(0.25), df['AGE'].quantile(0.75)
IQR    = Q3 - Q1
outliers = df[(df['AGE'] < Q1-1.5*IQR) | (df['AGE'] > Q3+1.5*IQR)]
print(f'Outliers: {len(outliers):,}')

plt.figure(figsize=(5, 4))
plt.boxplot(df['AGE'].dropna(), patch_artist=True,
            boxprops=dict(facecolor='lightblue', color='navy'))
plt.title('Age Boxplot — Outlier Detection')
plt.ylabel('Age')
plt.tight_layout()
plt.show()
```

```
Outliers: 37
```




```python
# 14. Hypothesis Test — Age vs Death (T-Test)
died_age     = df[df['DIED']==1]['AGE'].dropna()
survived_age = df[df['DIED']==0]['AGE'].dropna()
t_stat, p_val = stats.ttest_ind(died_age, survived_age)

print('T-Test: Age vs Death')
print(f'  Mean age (Died)    : {died_age.mean():.1f}')
print(f'  Mean age (Survived): {survived_age.mean():.1f}')
print(f'  P-value            : {p_val:.4e}')
print('  Result:', 'Reject H0 — Age significantly affects death' if
p_val < 0.05 else 'Fail to reject H0')
```

```
T-Test: Age vs Death
  Mean age (Died)    : 61.0
  Mean age (Survived): 51.4
  P-value            : 0.0000e+00
  Result: Reject H0 — Age significantly affects death
```

```python
# 15. EDA Summary
print('='*50)
print('  EDA SUMMARY — COVID-19 Mexico Dataset')
print('='*50)
print(f'Total Records : {len(df):,}')
print(f'Total Columns : {df.shape[1]}')
print(f'Total Deaths  : {df["DIED"].sum():,}')
print(f'Death Rate    : {df["DIED"].mean()*100:.2f}%')
print(f'Mean Age      : {df["AGE"].mean():.1f} years')
```

```python
print()
print('Key Insights:')
print('  - Older patients (60+) have higher death rate')
print('  - Diabetes & Hypertension are top comorbidities')
print('  - Pneumonia strongly linked to death')
print('  - Males have slightly higher death rate than females')
print('='*50)
```

```
==================================================
  EDA SUMMARY — COVID-19 Mexico Dataset
==================================================
Total Records : 236,526
Total Columns : 24
Total Deaths  : 30,103
Death Rate    : 12.73%
Mean Age      : 52.6 years

Key Insights:
  - Older patients (60+) have higher death rate
  - Diabetes & Hypertension are top comorbidities
  - Pneumonia strongly linked to death
  - Males have slightly higher death rate than females
==================================================
```